JDScience BTEC Level 3 Chemical Calculations Answer Sheet

Balanced equations, relative atomic mass and relative formula mass.

Mark scheme

Question	Answer / marking points
A1	$Mg + 2HCl \rightarrow MgCl_2 + H_2$
A2	$4Na + O_2 \rightarrow 2Na_2O$
A3	$Ca(OH)_2 + 2HNO_3 \rightarrow Ca(NO_3)_2 + 2H_2O$
A4	$4Al + 3O_2 \rightarrow 2Al_2O_3$
A5	$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$
A6	$2Fe + 3Cl_2 \rightarrow 2FeCl_3$
B	Ar = ((35 x 75)+(37 x 25))/100 = 35.5. It is a weighted mean because isotopes have different abundances.
C	CO_2=44; $Ca(OH)_2$=74; $Mg(NO_3)_2$=148; $Al_2(SO_4)_3$=342; $NaAl(OH)_4$=118
D	$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$. Same number of each atom on both sides, so atoms are conserved.

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